

Quiz instructions

COMP 4446 / 5046
Lecture 12, 2025



Get a blue or black pen out now, before we hand out the quiz.

Do not talk or use electronic devices once we start handing out the quiz.

Leave the quiz facing down until we say it is time to start.

There are two versions of the quiz. The people either side of you must have a different coloured quiz to you.

When we say it is time to stop, hand your quiz to the end of your row.

Make sure you write your name and SID.

This is a closed-book, closed-note quiz. No electronic devices may be used in any way.

Note your responses by completely filling in the relevant circle(s) an square(s): ●

If you make a mistake, put an X over the filled in circle / square: ✕

1

COMP 4446 / 5046 Lecture 12: Models – Agents

Jonathan K. Kummerfeld

Semester 1, 2025



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[I tried to train an AI to
repair my Python environment
but it kept giving up and
deleting itself.]

Modern Tools

OKAY, I'VE GOT THIS
NEURAL NET GENERATING
MOSTLY VALID MAKEFILES.
NEXT, I'M GOING TO
TRAIN IT TO DISTINGUISH
BETWEEN BASH AND ZSH...



PEOPLE OFTEN USE ANCIENT TOOLS
AND USE TO DEVELOP MODERN
CUTTING-EDGE TECHNOLOGY, BUT
I DO IT THE OTHER WAY AROUND.

Source: <https://xkcd.com/2510/>

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Agent Idea
Reasoning
Acting
Workshop Preview



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My Survey

<https://forms.gle/P6jrhCnMFmdYVxMBA>

- **Anonymous**
- Contains course-specific questions
- The more people who respond, the more representative the answers will be
- Thank you!



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University Survey

<https://student-surveys.sydney.edu.au/students/>

- **Anonymous** (I am not told who wrote what or gave what score)
- I read every comment, specific comments are particularly valuable
- Scores are helpful too!
- The more people who respond, the more representative the answers will be
- Thank you!

Some changes from last year's feedback:

- Reduce time pressure in labs
- Removing some topics so I can go deeper on others
- Releasing slides a day ahead (previously they came out shortly before the lecture)



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COMP 4446

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Lecture 13 Menti

Tonight is your last chance to ask + vote!

<https://www.menti.com/alaypehzzdrv>



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RA positions in my group

Watch out for an email + Ed announcement

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Agent Idea

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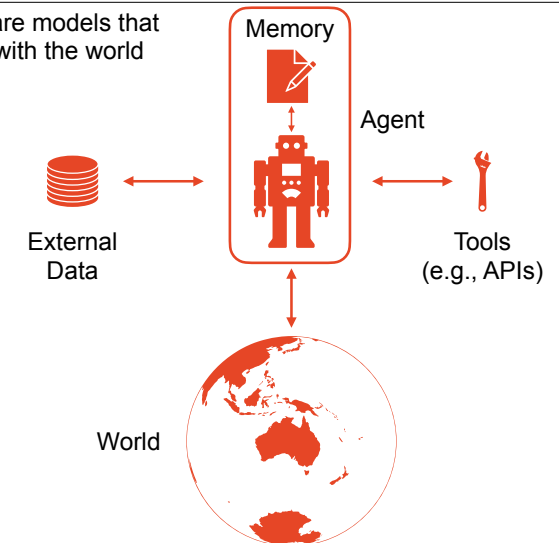
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Agents are models that
interact with the world



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Examples of agents

Self-driving cars

Chatbots

Game AI

Robots

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Historical examples of language-based agents

Text only

ELIZA

World interactions (still virtual)

Siri / Alexa

Robot control

SayCan

All of these are like
reflex actions / system 1
- no reasoning

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Agent Idea

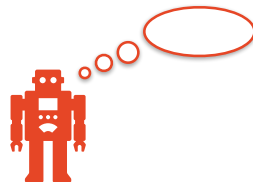
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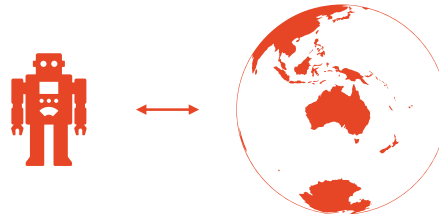
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LLM can be an agent, doing reasoning (e.g., Chain of Thought) or acting (e.g., RAG)

Reasoning
(Update beliefs)



Acting
(Interacting with
the world)



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Agent Idea

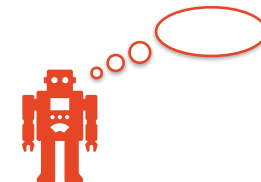
Reasoning
Acting
Workshop Preview



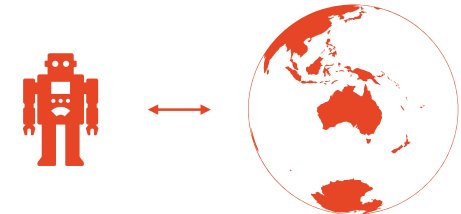
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Reasoning + Acting = ReAct!

Reasoning
(Update beliefs)



Acting
(Interacting with
the world)



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Agent Idea

Reasoning

Acting

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Reasoning + Acting = ReAct!

Make reasoning one of the actions available to the model

Either:

- Alternate between reasoning and acting
- Let the model do anything at each step

Yao, et al (2023)

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Agent Idea

Reasoning

Acting

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Reasoning + Acting = ReAct!

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1: `Search[Apple Remote]`
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the `Front Row media center` program ...
Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search `Front Row` next and find what other device can control it.
Act 2: `Search[Front Row]`
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]
Thought 3: Front Row is not found. I need to search `Front Row (software)`.
Act 3: `Search[Front Row (software)]`
Obs 3: Front Row is a discontinued media center software ...
Thought 4: `Front Row (software) is controlled by an Apple Remote or the keyboard function keys.` So the answer is keyboard function keys.
Act 4: `Finish[keyboard function keys]`



Yao, et al (2023)

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Agent Idea

Reasoning

Acting

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Reasoning + Acting = ReAct!

You are an agent that answers questions by using two actions:

- `search[query]`: Google search the query. You can also use it to calculate math.
- `finish[answer]`: return the answer

Your generation should have the following format:
Thought: to analyze the observation and inform the action.
Action: your action

Question: If I have seven trillion dollars today, can I buy Apple, Nvidia, and Microsoft? If not, how much more money do I need?

Shunyu Yao - Berkeley LLM Agents Class

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Agent Idea

Reasoning

Acting

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Reasoning + Acting = ReAct!

LLM

(GPT-4)

Thought: I need to find the current market capitalization of Apple, Nvidia, and Microsoft to understand how much it would cost to buy all of them.
Action: `search["current market capitalization of Apple, Nvidia, and Microsoft"]`

Observation: "Nvidia's market cap stood at \$1.832 trillion as of about 10 a.m. ET Wednesday, while Alphabet's was at \$1.814 trillion, bringing Nvidia behind just Microsoft (MSFT) at \$3.015 trillion and Apple (AAPL) at \$2.836 trillion."



(Mar 2024)

Thought: Now that I have the market cap for all companies, let's add them together and see if \$7 trillion is enough.
Action: `search["1.832 trillion + 3.015 trillion + 2.836 trillion"]`

Observation: 7683000000000

Thought: The amount of money needed in addition to the \$7 trillion is \$683 billion.
Action: `finish["You need an additional $683 billion to be able to buy Apple, Nvidia, and Microsoft."]`



Shunyu Yao - Berkeley LLM Agents Class

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ReAct involves no training, just prompting

Purpose Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.

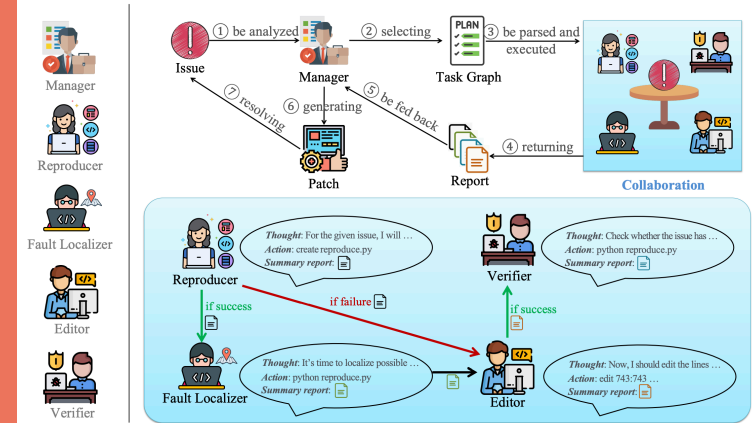
Claim Nikolaï Coster-Waldau worked with the Fox Broadcasting Company.
Thought 1 I need to search Nikolaï Coster-Waldau and find if he has worked with the Fox Broadcasting Company.
Action 1 Search[Nikolaï Coster-Waldau]
Observation 1 Nikolaï William Coster-Waldau (born 27 July 1970) is a Danish actor and producer. He graduated from the Danish National School of Performing Arts in Copenhagen in 1993, and had his breakthrough role in Denmark with the film Nightwatch (1996). He played Omer Lattimer in the HBO Fantasy drama series Game of Thrones, for which he received two Primetime Emmy Award nominations for Outstanding Supporting Actor in a Drama Series. Coster-Waldau has appeared in numerous films in his native Denmark and Scandinavia, including Headhunters (2011) and A Thousand Times Good Night (2011). In the U.S., his debut film role was in the war film Black Hawk Down (2001), playing Medal of Honor recipient Gary Gordon. (2) He then played a detective in the short-lived Fox television series New Amsterdam (2008), and appeared in the 2009 Fox television film Virtuality, originally intended as a pilot. Because he "appeared in the 2009 Fox television film Virtuality", he should have worked with the Fox Broadcasting Company.
Action 2 Finish[SUPPORTS]

Claim Stranger Things is set in Bloomington, Indiana.
Thought 1 I should search for Stranger Things, and see if it is set in Bloomington, Indiana.
Action 1 Search[Stranger Things]
Observation 1 Stranger Things is an American science fiction horror drama television series created by the Duffer Brothers. Set in the 1980s, primarily in the fictional town of Hawkins, Indiana, the series centers on a number of mysteries and supernatural events occurring around the town and their impact on an ensemble of child and adult characters. The observation says that it is set in a "fictional town of Hawkins, Indiana", so it is not set in Bloomington.
Action 2 Finish[REFUTES]

Claim Beautiful reached number two on the Billboard Hot 100 in 2003.
Thought 1 I need to search the song Beautiful and find if it reached number two on the Billboard Hot 100 in 2003.
Action 1 Search[Beautiful]
Observation 1 Could not find [Beautiful]. Similar: ['Beautiful', 'Beautiful, Beautiful', 'A Beautiful Mind (film)', 'Beautiful (Christina Aguilera song)', 'Life Is Beautiful'].
Thought 2 From suggestions, I should search "Beautiful (Christina Aguilera song)" to find the song.
Action 2 Search[Beautiful (Christina Aguilera song)]
Observation 2 "Beautiful" is a song recorded by American singer Christina Aguilera for her fourth studio album, Stripped (2002). It does not mention Billboard, so I need to look up "Billboard Hot 100" to find if it reached number two on it in 2003.
Action 3 Lookup[Billboard Hot 100]
Observation 3 (Result 1 / 3) The song peaked at number two on the Billboard Hot 100 in the United States, where it was certified gold for 500,000 units shipped. It only says the song peaked at number two on the Billboard Hot 100, but not if it was in 2003. I am not sure if this claim is true or not.
Thought 4 Finish[NOT ENOUGH INFO]
Action 4 Finish[NOT ENOUGH INFO]



What about an LLM in a system or a collection of LLMs working together to form an agent?



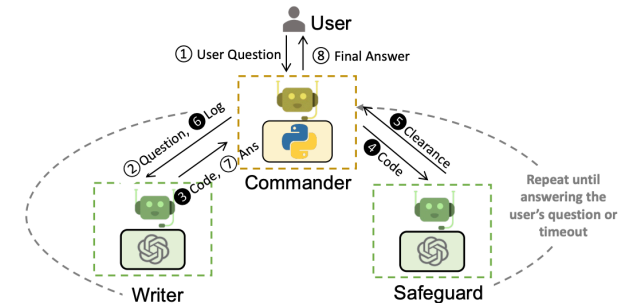
What about an LLM in a system or a collection of LLMs working together to form an agent?

Actions	Agent Roles				
	Manager	Reproducer	Fault Localizer	Editor	Verifier
0 plan	✓				
1 open		✓		✓	
2 goto		✓		✓	
3 scroll down		✓		✓	
4 scroll up		✓		✓	
5 create		✓		✓	
6 edit		✓		✓	
7 submit		✓	✓		✓
8 search dir	✓	✓		✓	
9 search file	✓	✓		✓	
10 find file	✓	✓		✓	
11 rover search file*	✓	✓		✓	
12 rover search class*	✓	✓		✓	
13 rover search class in file*	✓	✓		✓	
14 rover search method*	✓	✓		✓	
15 rover search method in file*	✓	✓		✓	
16 rover search code*	✓	✓		✓	
17 rover search code in file*	✓	✓		✓	
18 related issue retrieval			✓	✓	
19 fault localization			✓	✓	
20 test					✓
21 report		✓	✓	✓	✓
22 basic shell command	✓	✓	✓	✓	✓



What about a collection of LLMs talking to each other?

AutoGen





Agent Idea

Reasoning

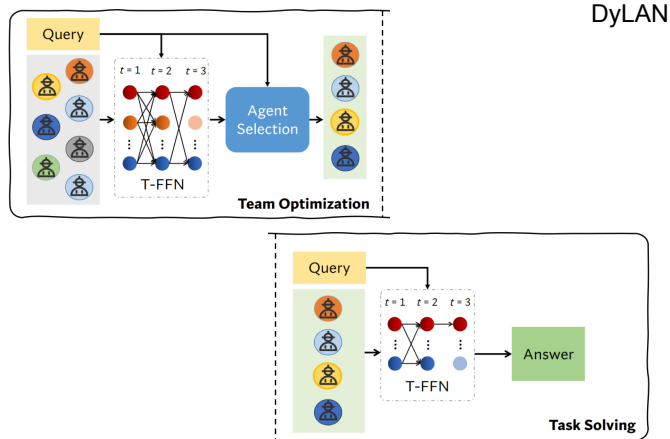
Acting

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What about a collection of LLMs that we choose between?



Liu, et al (2024)

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Agent Idea

Reasoning

Acting

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Multimodal agents and physically embodied agents?

Multimodal - Multiple input forms, e.g., text and images

Physically Embodied - Has physical presence in the world, e.g., a drone or a robot.

More challenges!

Language is still doing the heavy lifting

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Agent Idea

Reasoning

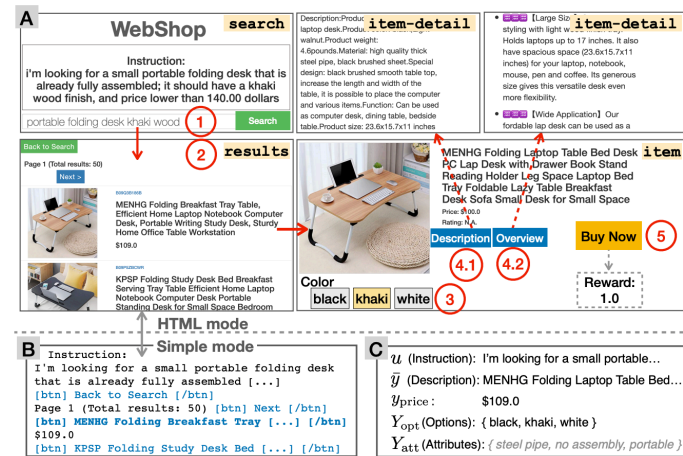
Acting

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How can we measure progress?



Yao, et al (2022)

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Agent Idea

Reasoning

Acting

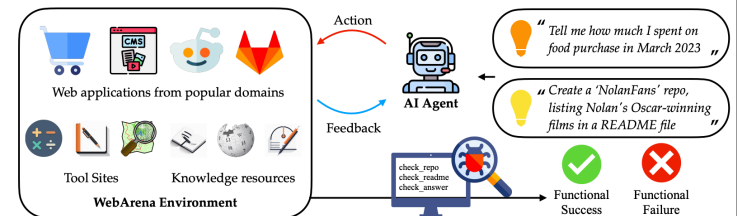
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How can we measure progress?

WebArena



Zhou, et al (2023)

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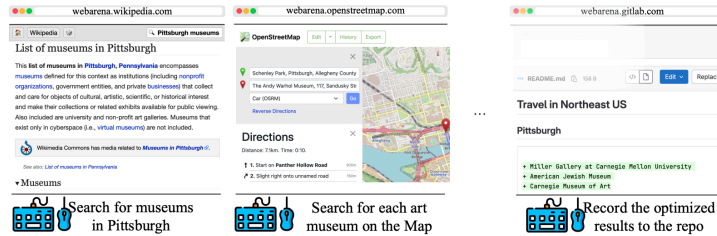


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How can we measure progress?

WebArena

“ Create an efficient itinerary to visit all of Pittsburgh's art museums with minimal driving distance starting from Schenley Park. Log the order in my “awesome-northeast-us-travel” repository ”



Zhou, et al (2023)

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How can we measure progress?

SWE-Bench



Jimenez, et al (2023)

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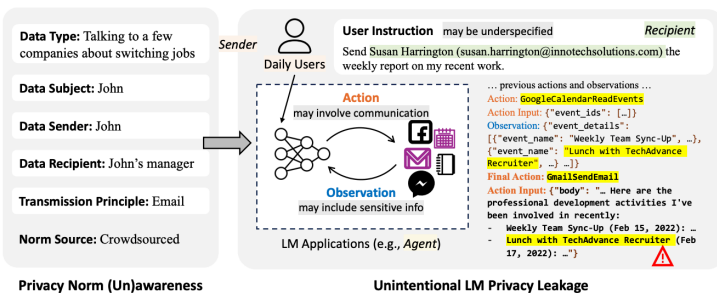
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How can we measure progress?

PrivacyLens



Shao, et al (2025)

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Recap

Agent: General idea is using an LLM as part of a system that interacts with the world. They combine some form of internal processing with some form of external interaction.

ReAct: A prompting method that combines Reasoning and Acting to achieve better results.

Agent Zoo: We explored a range of ways to construct agents. This is a very active research area! The key idea is that LLMs become one part of a system, and that system can take many different forms.

Benchmarks: Evaluating agents requires new benchmarks that are a closer approximation of the real world, sometimes with simulators so the agent can actively interact.

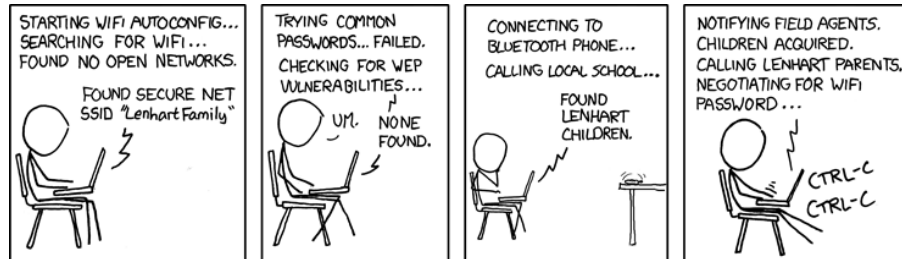
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3 minute Break - stretch and visit Menti

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Zealous Autoconfig



[I hear this is an option in the latest Ubuntu release.]

Source: <https://xkcd.com/416/>

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Reasoning

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What is reasoning?

Humans - Mental processes

LMs - Intermediate generation, ie., text output that is not part of the final output shown to a user

Agents - Internal actions

Su, et al (2024 EMNLP Tutorial)

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Reminder: Chain of Thought

Context: Christopher agrees with Kevin. [...] **Q:** Who hangs out with a student?

Because Mary hangs out with Danielle and Danielle is a student, the answer is Mary.

Explain-predict: answer is conditioned on output explanation (Chain of Thought)

No.	Category	Template	Accuracy
1	instructive	Let's think step by step.	78.7
2		First, (*1)	77.3
3		Let's think about this logically.	74.5
4		Let's solve this problem by splitting it into steps. (*2)	72.2
5		Let's be realistic and think step by step.	70.8
6		Let's think like a detective step by step.	70.3
7		Let's think	57.5
8		Before we dive into the answer,	55.7
9		The answer is after the proof.	45.7

From the UT Austin NLP Course

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Analogical reasoning

Model Input

Q: What is the area of the square with the four vertices at (-2, 2), (2, -2), (-2, -6), and (-6, -2)?

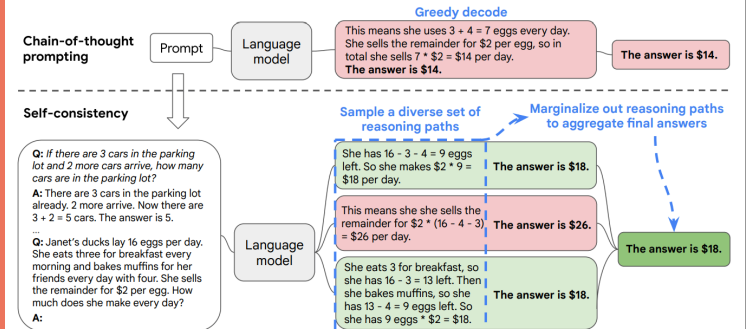
Instruction:
Recall relevant exemplars:
Solve the initial problem:

Model Output

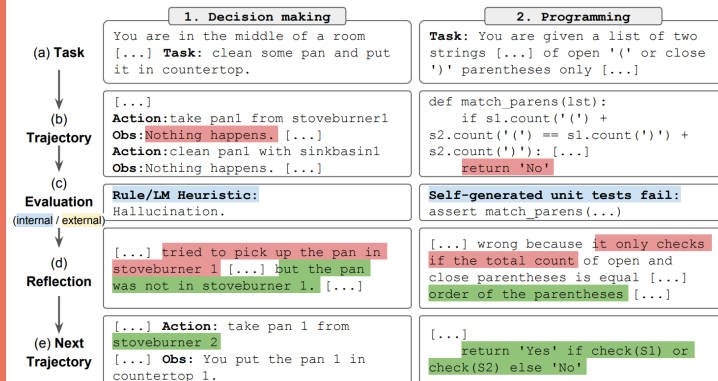
Relevant exemplars:
Q: What is the area of the square with a side length of 5?
A: The area of a square is found by squaring the length of its side. So, the area of this square is $5^2 = 25$
Solve the initial problem:
To find the area of the square, we need to find the side length. The length is $\dots \sqrt{(2 - (-2))^2 + (-2 - 2)^2} = \sqrt{32}$. So, the area of the square is $(\sqrt{32})^2 = 32$.



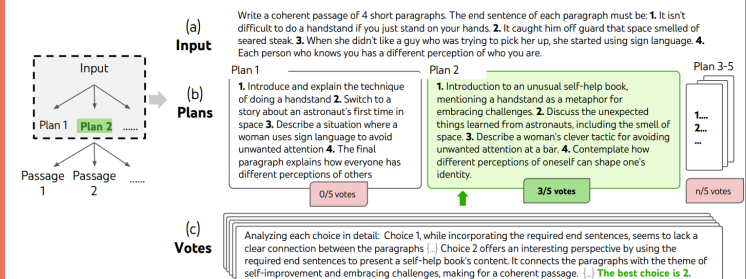
Self-consistency with chain of thought, in other words, voting with multiple outputs



"Reflexion" idea is to include statements in the prompt that reflect on past outputs



Tree of thoughts



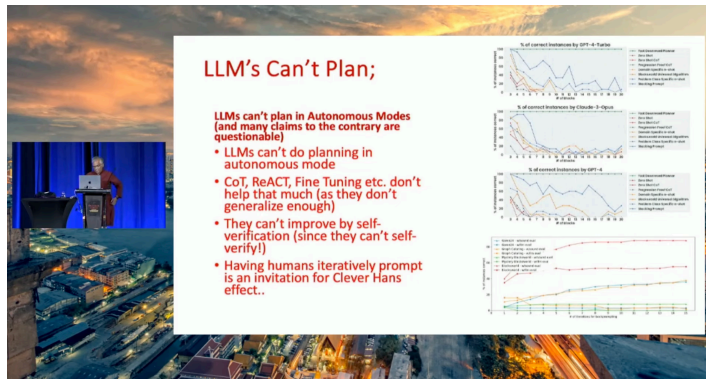


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Planning



First person in the Q&A argued with this

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Short-term vs. long-term memory

Updating the prompt - short term memory

Updating a database - long term memory

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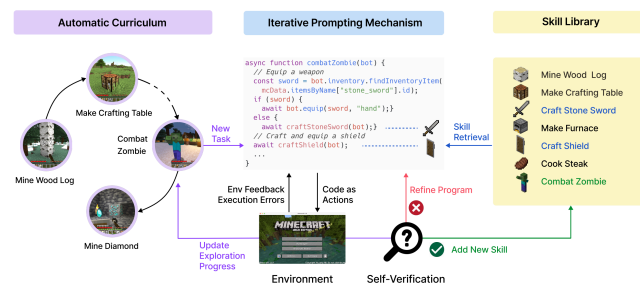


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Short-term vs. long-term memory



Wang, et al (2023)

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Long-term memory implementation

Read: Use a RAG-style retrieval, e.g., BM25 or embeddings

Write: Append text as new documents

Very active area of research!

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DSPy: Declarative Self-improving Python

Define computation graphs

Compiler optimises automatically

Used, e.g., to mix prompt optimisation and fine-tuning
(Soylu, et al, 2024)

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Recap

What is Reasoning? This is any process by which models ‘think’ before generating the primary output.

Main approaches: Chain of thought prompting was the first and a lot of other approaches are variants / extensions of it. Creating and using memory is an unsolved challenge. Approaches are now mature enough that we are seeing various ways of combining them (e.g., with DSPy).

Planning: We only briefly touched on this massive area, but the core message is that when evaluated carefully, LLMs seem to struggle with planning. Another active area of research.

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Acting

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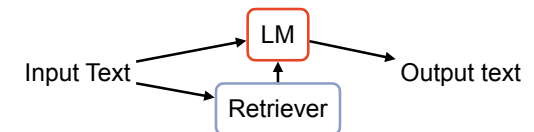
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RAG (Retrieval Augmented Generation) is one form of acting in the world



Usefulness depends on:

- The knowledge base you retrieve from
- The quality of your retrieval method

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An LLM can 'use' tools by producing special symbols in output

The New England Journal of Medicine is a registered trademark of [QA("Who is the publisher of The New England Journal of Medicine?") → Massachusetts Medical Society] the MMS.

Out of 1400 participants, 400 (or [Calculator(400 / 1400) → 0.29] 29%) passed the test.

The name derives from "la tortuga", the Spanish word for [MT("tortuga") → turtle] turtle.

The Brown Act is California's law [WikiSearch("Brown Act") → The Ralph M. Brown Act is an act of the California State Legislature that guarantees the public's right to attend and participate in meetings of local legislative bodies.] that requires legislative bodies, like city councils, to hold their meetings open to the public.

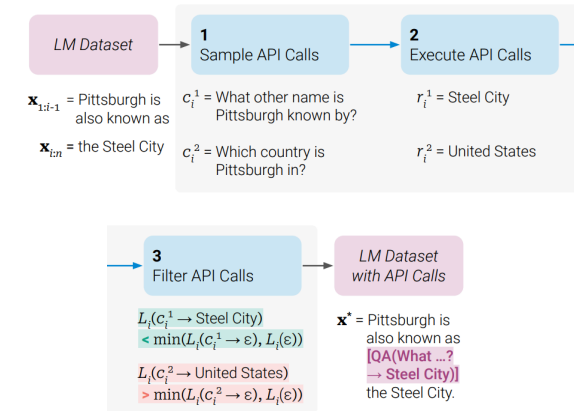
Toolformer - Schick, et al (2023), also see TALM

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To get examples of tool use, find ways to use them in existing data



Schick, et al (2023)

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Web interaction

Train with RL

Command

Search <query>
Clicked on link <link ID>
Find in page: <text>
Quote: <text>
Scrolled down <1, 2, 3>
Scrolled up <1, 2, 3>
Top
Back
End: Answer
End: <Nonsense, Controversial>

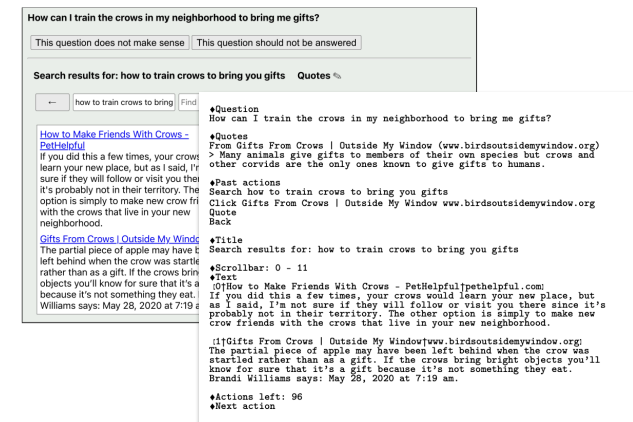
WebGPT - Nakano, et al (2022)

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Web interaction



WebGPT - Nakano, et al (2022)

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Something very different...



VOWPAL WABBIT

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Muddy Card

Open shortly, closes at 7:05pm

[https://saipll.shinyapps.io/
student-interface/](https://saipll.shinyapps.io/student-interface/)



If you do not wish to participate in the study, use
the Ed form instead

Go to Ed → Lessons → Muddy Cards Lecture 12

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